ARTIFICIAL INTELLIGENCE APPROACHES FOR PERSONALIZED NUTRITION, FITNESS, MENSTRUAL HEALTH AND COSMETIC ANALYSIS: A TECHNIQUE-BASED SURVEY

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***Abstract*—Artificial Intelligence (AI) and Machine Learning (ML) are increasingly being applied to personalized healthcare for prediction, recommendation, monitoring, and automated assessment. Recent research has explored AI techniques in personalized nutrition, fitness, menstrual health, cosmetic safety, skin-tone analysis, and dermatology. This survey reviews twelve selected studies with emphasis on their techniques, datasets, evaluation methods, results, and limitations. The reviewed approaches include rule-based nutrition recommendation [3], machine-learning-based workout recommendation [2], probabilistic menstrual-cycle prediction [6], Naive Bayes for cosmetic sensitization [5], hybrid deep learning for menstrual nutrition [10], and computer-vision-based skin-tone analysis [7]. Systematic reviews and meta-analyses related to exercise performance and muscle damage are also considered [4], [11]. The comparison shows that AI methods can achieve promising results for specific healthcare tasks, but their performance depends on dataset quality, population diversity, validation, and application context. Common challenges include small or simulated datasets, heterogeneous evaluation methods, limited real-world validation, demographic bias, privacy concerns, and limited interpretability. Future research should emphasize diverse datasets, standardized evaluation, explainable AI, privacy preservation, and longitudinal validation.**

***Index Terms*—Artificial Intelligence, Machine Learning, Deep Learning, Personalized Healthcare, Nutrition, Fitness, Menstrual Health, Cosmetic Analysis, Skin Tone.**

I. Introduction

Artificial Intelligence has become an important technology in healthcare because it can process large amounts of data and identify patterns that may be difficult to recognize manually. Machine learning and deep learning are increasingly used for prediction, classification, recommendation, monitoring, and automated assessment. The availability of wearable devices, digital health applications, medical images, and large datasets has expanded the use of AI in personalized healthcare [2], [6], [7].

Personalized healthcare considers individual differences rather than providing identical recommendations to all users. Dietary requirements, physical activity, menstrual characteristics, symptoms, and skin properties can vary considerably between individuals. Consequently, researchers have investigated AI-based methods for personalized nutrition, exercise recommendation, menstrual-cycle prediction, cosmetic safety, and skin analysis [1], [3], [5], [7].

Nutrition research includes both rule-based and deep-learning approaches. A rule-based system used meal filtering and scoring to match dietary requirements [3], while a hybrid deep-learning method combined feature extraction, classification, and optimization for nutrition recommendation during the menstrual cycle [10]. Fitness research has investigated smartwatch-assisted ML recommendations as well as the relationship between menstrual-cycle phase and exercise performance [1], [2].

Menstrual-health research has focused on cycle prediction, exercise performance, muscle damage, and evaluation of mobile applications [4], [6], [11], [12]. AI has also been applied to cosmetic ingredient sensitization using Naive Bayes [5] and to skin-tone analysis using CIELAB values and computer-vision landmarks [7]. Dermatological research has further explored CNNs and transfer learning for image classification and segmentation [9].

This survey groups the reviewed studies according to their application areas and AI techniques and compares their data, performance, limitations, and future research requirements.

II. Background of AI-Based Personalized Healthcare

A. Personalized Nutrition

Personalized nutrition aims to recommend foods according to nutritional requirements, dietary restrictions, allergies, preferences, and other individual characteristics. AI can perform filtering, scoring, classification, and recommendation based on such information [3], [10].

A rule-based AI nutrition recommendation system used 180 expert-validated Mediterranean meals and 4,000 simulated user profiles [3]. The system considered allergies, preferences, seasonal availability, calorie requirements, and nutritional targets. It achieved 100% filtering accuracy for the specified constraints, approximately 90% meal-plan completeness, and

98% caloric matching [3]. However, the use of simulated users and Mediterranean-focused data limits generalization.

A hybrid deep-learning approach for women's nutrition during the menstrual cycle used LSAENet for feature extraction, HABi-ConGRNet for classification, and Greylag Goose Optimization [10]. Using 2,000 samples, it achieved 97.52% accuracy, 97.82% precision, and 96.91% recall. Although performance was high, the architecture was relatively complex and used a specialized dataset [10].

B. Fitness and Exercise

AI-based fitness systems can analyze exercise and wearable-device information for personalized exercise prescription and monitoring. A systematic review investigated ML algorithms for smartwatch-assisted exercise recommendation [2]. The reviewed studies used different models, datasets, and evaluation measures, making direct comparison difficult. Limited long-term real-world validation was also identified as a major limitation [2].

The effect of menstrual-cycle phase on resistance training was investigated in a prospective longitudinal study involving 28 resistance-trained women [1]. Cycle phases were verified using multiple methods, and linear mixed-effects models were used for analysis. The study reported phase-related effects for some exercises and associations involving motivation and pain [1].

C. Menstrual Health

AI-based menstrual-health research includes cycle prediction and digital health application evaluation. A predictive model using data from more than 186,000 users incorporated adherence behavior in menstrual self-tracking [6]. It was evaluated using RMSE, absolute error, median absolute error, and cycle-length difference and outperformed several baseline approaches, including mean, median, CNN, RNN, and LSTM models [6].

A study evaluating 14 menstrual-health applications used an adapted mMARS framework to examine functionality, inclusiveness, and health education [12]. All applications provided cycle and symptom tracking, while 71.4% were reported to share data with third parties and 42.9% cited medical literature [12]. The study also identified limited validated symptom tools and healthcare-professional involvement.

D. Menstrual Cycle and Exercise

Systematic reviews have investigated whether menstrual-cycle phase influences exercise performance and exercise-induced muscle damage. A meta-analysis of 19 studies reported significant differences in delayed-onset muscle soreness and strength loss but not creatine kinase [4]. Variation in exercise protocols and phase-verification methods limited comparison between studies.

A Bayesian meta-analysis covering 78 studies and 1,193 women found that overall menstrual-cycle effects on exercise performance were generally trivial, with low-quality evidence [11]. Inconsistent phase verification and methodological variation were important limitations [11].

E. Cosmetic and Skin Analysis

Machine learning has been used to predict the sensitizing potential of cosmetic ingredients. A Naive Bayes model using physicochemical descriptors achieved 86% accuracy, 80% sensitivity, and 90% specificity [5]. The approach demonstrates the usefulness of classical ML for specialized cosmetic-safety classification but has a narrow application scope.

Computer vision has also been applied to automated skin-tone assessment. One study used CIELAB values obtained from DensePose and OpenFace landmarks and evaluated 214 clinical 3D scans and 551 AI-generated images [7]. Monk-scale classification achieved approximately 89–92% accuracy on clinical data. However, differences between clinical and generated images and limited representation of some skin tones remain concerns [7].

In dermatology, CNNs, transfer learning, and architectures such as ResNet, DenseNet, EfficientNet, and U-Net have been investigated for image classification and lesion segmentation [9]. Although strong results have been reported, demographic and device-related bias and limited prospective clinical validation remain important challenges [9].

III. AI Techniques and Common Features

A. Rule-Based Recommendation

Rule-based systems use predefined conditions to generate decisions or recommendations. The nutrition recommendation system in [3] used filtering and scoring rules based on dietary constraints and nutritional requirements. Such systems are simple, transparent, and easy to understand. However, they depend on manually defined rules and may not adapt well to changing user characteristics [3].

B. Classical Machine Learning

Classical ML remains effective for structured classification problems. The cosmetic sensitization study used Naive Bayes with physicochemical descriptors and achieved 86% accuracy [5]. The main advantage of classical ML is relatively low computational complexity compared with deep-learning systems. Performance, however, depends strongly on the quality and relevance of input features.

C. Probabilistic Machine Learning

Probabilistic methods are useful for prediction tasks involving individual variation and uncertainty. The menstrual-cycle prediction model incorporated adherence behavior and used data from more than 186,000 users [6]. Its performance exceeded several conventional and deep-learning baselines, demonstrating the value of individualized prediction.

D. Deep Learning

Deep learning can learn complex patterns from large datasets and is widely used for image analysis and complex classification. The hybrid nutrition method in [10] combined feature extraction, classification, and optimization and achieved more than 97% accuracy, precision, and recall.

CNN-based approaches are also common in dermatology. ResNet, DenseNet, EfficientNet, and U-Net have been applied to image classification and segmentation [9]. Transfer learning can improve performance when task-specific datasets are limited. However, deep models can be computationally expensive and difficult to interpret [9], [10].

E. Computer Vision

Computer vision enables automatic extraction of information from images. The skin-tone analysis study combined CIELAB color representation with DensePose and OpenFace landmarks [7]. This demonstrates how image-processing and AI techniques can be combined for quantitative skin assessment. Image quality, lighting, demographic representation, and acquisition devices can affect performance.

F. Optimization

Optimization algorithms can be combined with deep-learning models to improve model parameters or feature selection. The Greylag Goose Optimization method used in [10] was combined with LSAENet and HABi-ConGRNet to improve nutrition classification performance. However, additional optimization stages can increase computational and implementation complexity.

IV. Comparative Analysis

The reviewed studies differ in application, data type, methodology, sample size, and evaluation criteria. Table I summarizes the major characteristics.

The comparison shows that no single AI technique is suitable for all personalized-healthcare tasks. Rule-based systems provide transparency and simplicity [3], while classical ML methods are useful for structured classification [5]. Probabilistic ML is suitable for prediction involving individual variation, such as menstrual-cycle prediction [6].

Deep-learning approaches can achieve high performance for complex nutrition and image-analysis tasks [9], [10], but they generally require suitable datasets and may be difficult to interpret. Computer-vision methods are particularly useful for image-based skin analysis, although performance can be influenced by image quality and demographic representation [7].

The reviewed datasets vary substantially. The cycle-prediction study used more than 186,000 users [6], while the hybrid nutrition model used 2,000 samples [10] and the exercise-performance study included only 28 participants [1]. Therefore, reported accuracy values cannot be directly compared across studies because the tasks, datasets, and evaluation procedures differ.

Another important observation is that high performance on a specialized dataset does not necessarily indicate broad generalizability. For example, the nutrition recommendation study used simulated user profiles [3], while the skin-tone study included both clinical and AI-generated images [7]. External validation using diverse real-world populations is therefore necessary.

V. Research Gaps and Future Directions

A. Dataset Diversity

Small, simulated, or specialized datasets are common limitations in the reviewed studies [1], [3], [7], [10]. Such datasets may not represent the diversity of real users. Future research should include larger and more diverse populations covering differences in age, geographical background, dietary habits, activity levels, menstrual characteristics, and skin tones.

B. Real-World Validation

Limited longitudinal and real-world validation remains a major concern [2], [9]. Models should be evaluated over longer periods and in realistic environments to determine whether their performance remains reliable outside controlled datasets.

C. Standardized Evaluation

The reviewed studies use different datasets, metrics, validation strategies, and experimental designs [2], [4], [11]. Standardized evaluation protocols would improve reproducibility and make comparisons between AI approaches more meaningful.

D. Explainability

Complex deep-learning and hybrid models can be difficult to interpret [9], [10]. Explainable AI methods should therefore be incorporated to provide understandable reasons for predictions and recommendations, particularly in healthcare applications.

E. Bias and Fairness

Demographic bias is an important concern in AI-based healthcare. Skin-tone analysis has shown differences related to representation and image type [7], while dermatological AI can be affected by demographic and device-related bias [9]. Future datasets should provide balanced representation and performance should be assessed across population groups.

F. Privacy and Security

Personalized healthcare applications may process sensitive health, menstrual, activity, dietary, and image data. The evaluation of menstrual-health applications reported that 71.4% shared data with third parties [12]. Privacy-preserving ML, secure storage, transparent data policies, and informed consent should therefore receive greater attention.

G. Clinical Validation and Usability

Several reviewed approaches have limited involvement of healthcare professionals or prospective clinical validation [9], [12]. Future systems should be evaluated with appropriate professional input and realistic users. Model complexity should also be balanced against computational requirements and practical usability [10].

TABLE I
COMPARISON OF REVIEWED PAPERS

Paper No.	Application	Technique	Dataset / Sample	Main Result	Major Limitation
[1]	Exercise performance	Longitudinal analysis + linear mixed-effects models	28 resistance-trained women	Phase-related effects in some exercises	Small and limited sample
[2]	Workout recommendation	ML + smartwatch-assisted exercise prescription	Human studies from 5 databases	Potential for personalized exercise recommendation	Heterogeneous studies and limited long-term validation
[3]	Personalized nutrition	Rule-based filtering and scoring	180 meals + 4,000 simulated profiles	98% caloric matching	Simulated users and limited cuisine diversity
[4]	Muscle damage	Systematic review + meta-analysis	19 studies	Differences in DOMS and strength loss	Protocol and phase-verification variation
[5]	Cosmetic safety	Naive Bayes + physico-chemical descriptors	Cosmetic ingredient data	86% accuracy; 80% sensitivity; 90% specificity	Limited to sensitization prediction
[6]	Cycle prediction	Probabilistic ML	>186,000 users	Outperformed several baseline models	Limited to cycle-start prediction
[7]	Skin-tone analysis	CIELAB + DensePose/OpenFace	214 clinical scans + 551 generated images	89–92% Monk-scale accuracy	Skin-tone imbalance and limited validation
[8]	Healthcare translation	AI-based real-time translation	9 bilingual clinicians	Non-inferiority in selected domains	Small scripted study
[9]	Dermatology	CNN + transfer learning + segmentation	Dermatological images	Strong classification and segmentation results	Bias and limited clinical validation
[10]	Menstrual nutrition	Hybrid DL + Greylag Goose Optimization	2,000 samples	97.52% accuracy; 97.82% precision; 96.91% recall	Complex architecture and specialized data
[11]	Exercise performance	Systematic review + Bayesian meta-analysis	78 studies; 1,193 women	Overall effects generally trivial	Low-quality evidence and methodological variation
[12]	Menstrual-health apps	Adapted mMARS evaluation	14 applications	100% provided cycle/symptom tracking	Limited validated tools and professional involvement

VI. CONCLUSION

This survey reviewed twelve studies covering AI applications in personalized nutrition, fitness, menstrual health, cosmetic safety, skin-tone analysis, and dermatology. The literature demonstrates the use of rule-based systems, classical machine learning, probabilistic prediction, deep learning, computer vision, optimization, and systematic evidence synthesis [2]–[7], [9]–[12].

Rule-based methods provide transparent recommendations [3], while classical ML can effectively handle structured classification tasks such as cosmetic sensitization [5]. Probabilistic ML has shown potential for large-scale menstrual-cycle prediction [6]. Deep-learning and optimization-based approaches have reported high performance in nutrition and dermatological applications [9], [10]. Computer-vision techniques also provide promising approaches for automated skin-tone analysis [7].

However, the reviewed literature highlights several limitations, including small or specialized datasets, simulated data, inconsistent evaluation, limited longitudinal validation, demographic bias, privacy concerns, and limited interpretability [1]–[3], [7], [9], [10], [12]. Therefore, high predictive accuracy alone is not sufficient for reliable healthcare AI.

Future research should focus on diverse datasets, standardized evaluation, external and longitudinal validation, explainable AI, privacy preservation, fairness, and stronger clinical involvement. These improvements are essential for developing reliable and generalizable AI approaches for personalized healthcare.

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